Internet of Things

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1) Changing LED with Phone

ESP LED BL 1

```
#include "BluetoothSerial.h"
BluetoothSerial SerialBT;
int received;
char receivedChar;
const char turnRed ='r';
const char turnGreen ='g';
const char turnBlue ='b';
const char turnOFF ='f';
const int redpin = 21;
const int bluepin = 22;
const int greenpin = 23;
int LEDpin;
void setup() {
Serial.begin(115200);
SerialBT.begin("ESP32_IOTLab"); //Bluetooth device name
Serial.println("The bluetooth device is ready to pair!");
Serial.println("To turn RED send: r");
Serial.println("To turn GREEN send: g");
Serial.println("To turn BLUE send: b");
Serial.println("To turn OFF send: f");
pinMode(redpin, OUTPUT);
pinMode(greenpin, OUTPUT);
pinMode(bluepin, OUTPUT);
pinMode(LEDpin, OUTPUT);
}
```

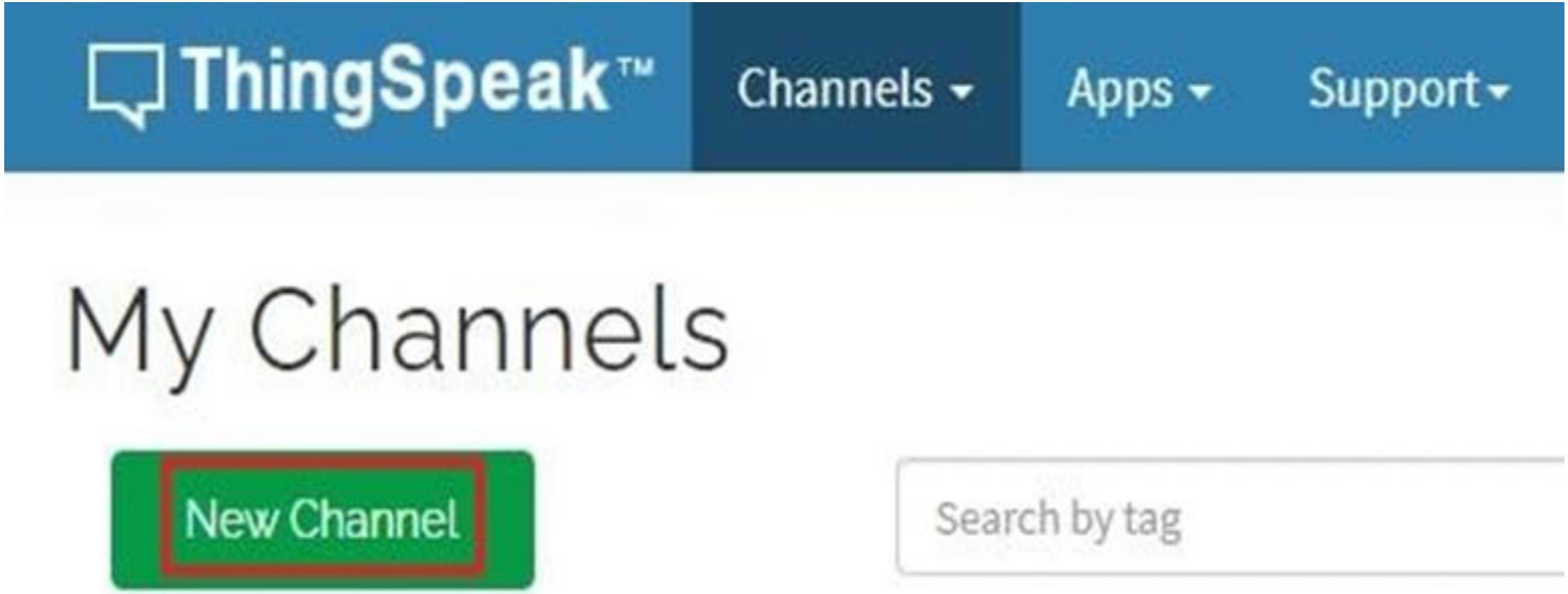
```
void loop() {
receivedChar =(char)SerialBT.read();
if (Serial.available())
{
SerialBT.write(Serial.read());
}
if (SerialBT.available())
{
SerialBT.print("Received:");
SerialBT.println(receivedChar);
Serial.print ("Received:");
Serial.println(receivedChar);
if(receivedChar == turnRed)
{
SerialBT.println("LED ON:");
Serial.println("LED ON:");
digitalWrite(redpin, HIGH);
digitalWrite(greenpin, LOW);
digitalWrite(bluepin, LOW);
LEDpin = redpin;
}
if(receivedChar == turnGreen)
{
digitalWrite(greenpin, HIGH);
digitalWrite(redpin, LOW);
digitalWrite(bluepin, LOW);
LEDpin = greenpin;
}
if(receivedChar == turnBlue)
{
SerialBT.println("LED ON:");
Serial.println("LED ON:");
digitalWrite(bluepin, HIGH);
digitalWrite(redpin, LOW);
digitalWrite(greenpin, LOW);
LEDpin = bluepin;
}
if(receivedChar == turnOFF)
{
SerialBT.println("LED OFF:");
Serial.println("LED OFF:");
digitalWrite(LEDpin, LOW);
}
}
delay(20);
}
```

2) DHT with Thinkspeak

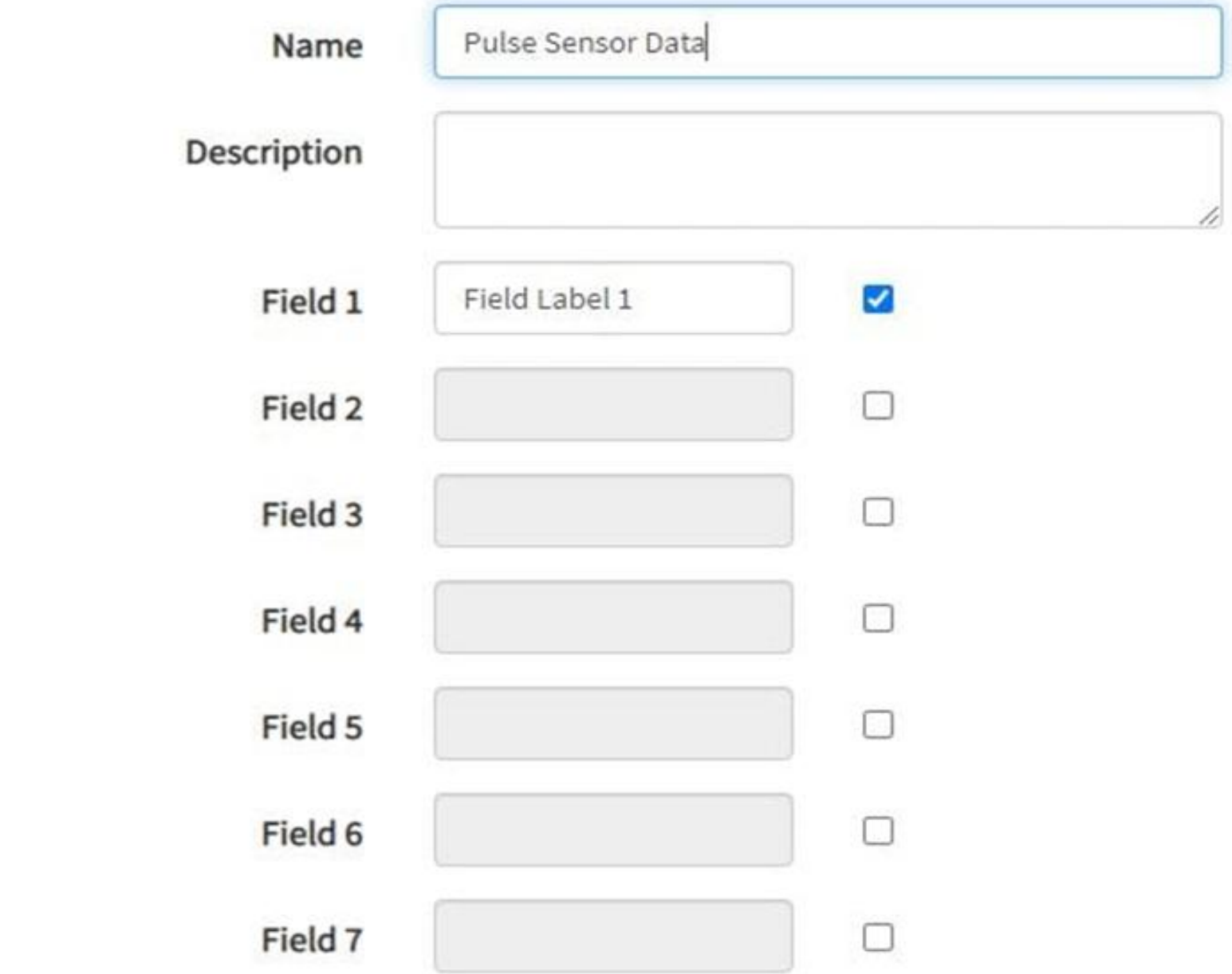
Thingspeak

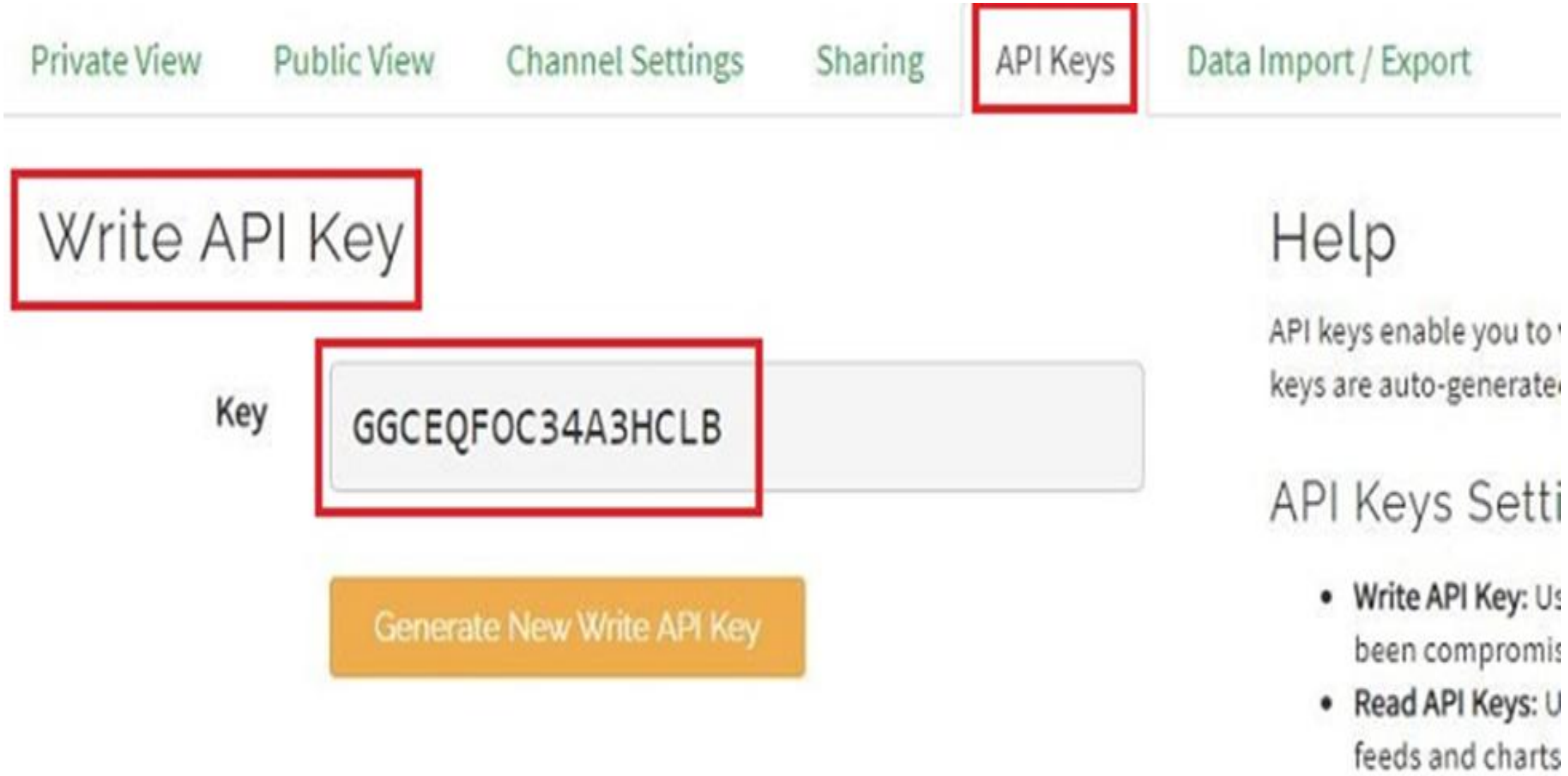
ThingSpeak™ is an IoT analytics platform service that allows you to aggregate, visualize and analyze live data streams in the cloud. ThingSpeak provides instant visualizations of data posted by your devices to ThingSpeak. With the ability to execute MATLAB® code in ThingSpeak you can perform online analysis and processing of the data as it comes in. ThingSpeak is often used for prototyping and proof of concept IoT systems that require analytics.

Contd.


ThingSpeak™
Channels
Apps
Support
My Channels
New Channel
Search by tag

New Channel


Name
Pulse Sensor Data
Description
Field 1
Field Label 1
Field 2
Field 3
Field 4
Field 5
Field 6
Field 7


Private View
Public View
Channel Settings
Sharing
API Keys
Data Import / Export
Write API Key
Key
GGCEQFOC34A3HCLB
Generate New Write API Key
Help
API keys enable you to
keys are auto-generate
API Keys Setti
Write API Key: Us
been compromis
Read API Keys: U
feeds and charts

Code Link

https://github.com/vipulhld001/IoT-Lab-Code/

MQTT using Mosquitto

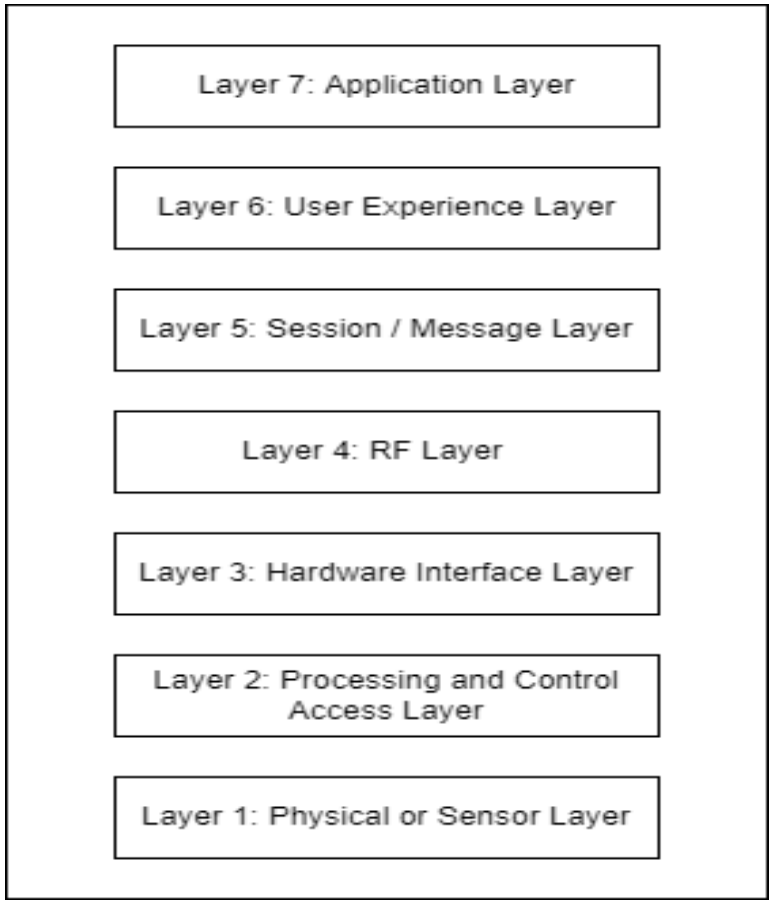
Outline

- Introduction.
- IoT Framework.
- What is MQTT?
- Characteristics of MQTT.
- MQTT Architecture.
- Eclipse Mosquitto.
- Basics of Mosquitto.
- Sub and Pub.
- To Publish.
- To Subscribe.
- Filtering.

Introduction

- Protocols act as the backbone of IoT.
- Messaging protocols are vital in sending and receiving the data/message to /from the cloud for any IoT application.
- Two major protocols in this regard are: –
  - MQTT (Message Queuing Telemetry Transport)
  - CoAP (Constrained Application Protocol)

IoT Framework



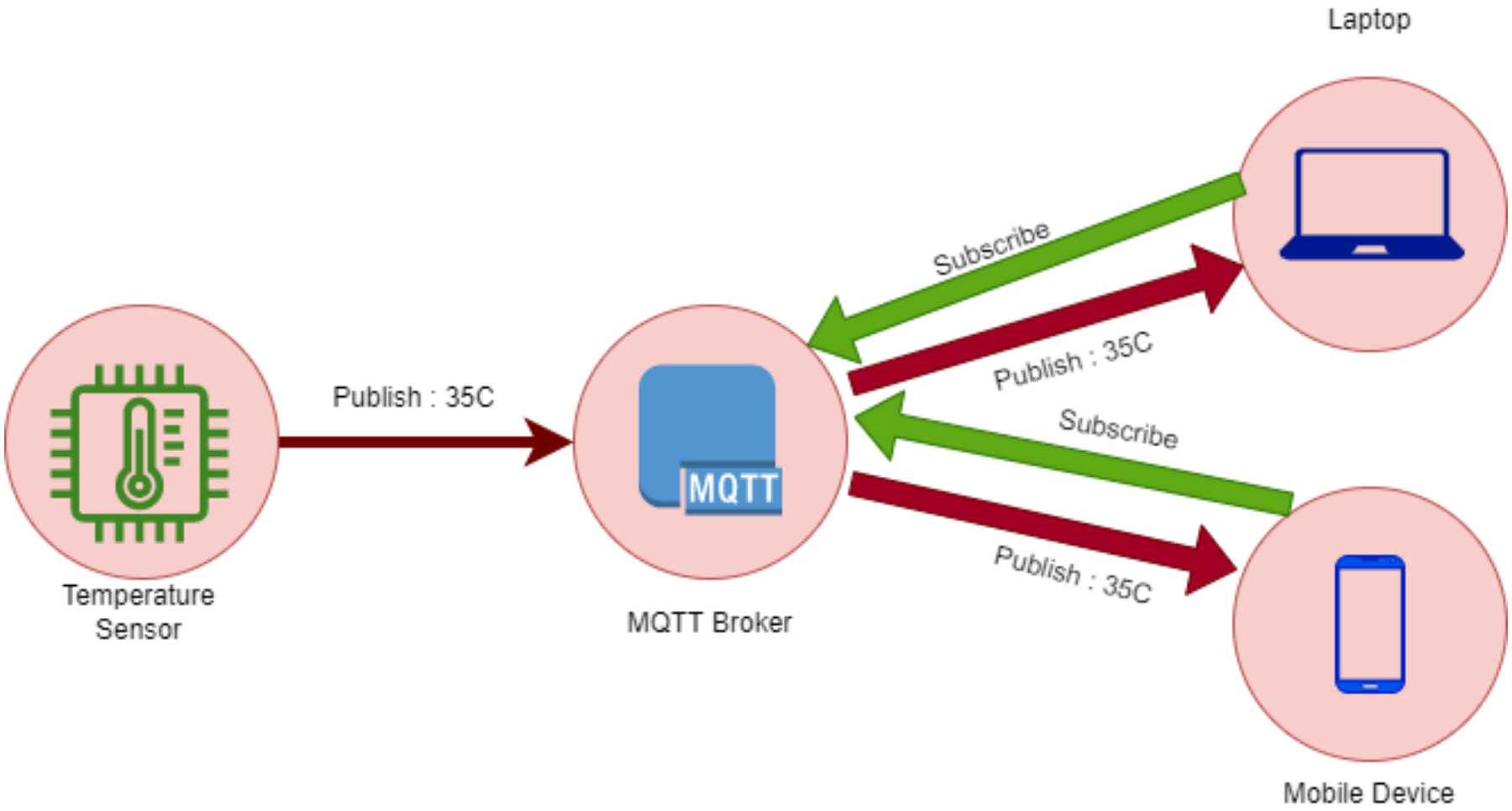
MQTT is a part of Layer 5

What is MQTT?

- MQTT was designed by Andy Stanford Clark & Arlen for IBM 1999 as a proprietary protocol. By 2014, this was made free and open. This began the revolution in IoT.
- MQTT follows the publish-subscribe pattern.
- Thus, a central broker plays a key role in the entire system.
- This message broker helps to dispatch messages to the nodes that have subscribed for the same.
- The publisher node sends messages to the broker, and the broker takes responsibility for dispatching the messages to the subscribed destinations.

Contd.

- MQTT is a lightweight protocol, which makes it exceptionally useful.
- Lightweight means that it demands minimal resources for its functioning and does not require additional resources from its working environment.
- IoT prefers this type of lightweight protocol due to resource constraints.


Laptop
Subscribe
Publish : 35C
Publish : 35C
Subscribe
Publish : 35C
MQTT
Temperature Sensor
MQTT Broker
Mobile Device

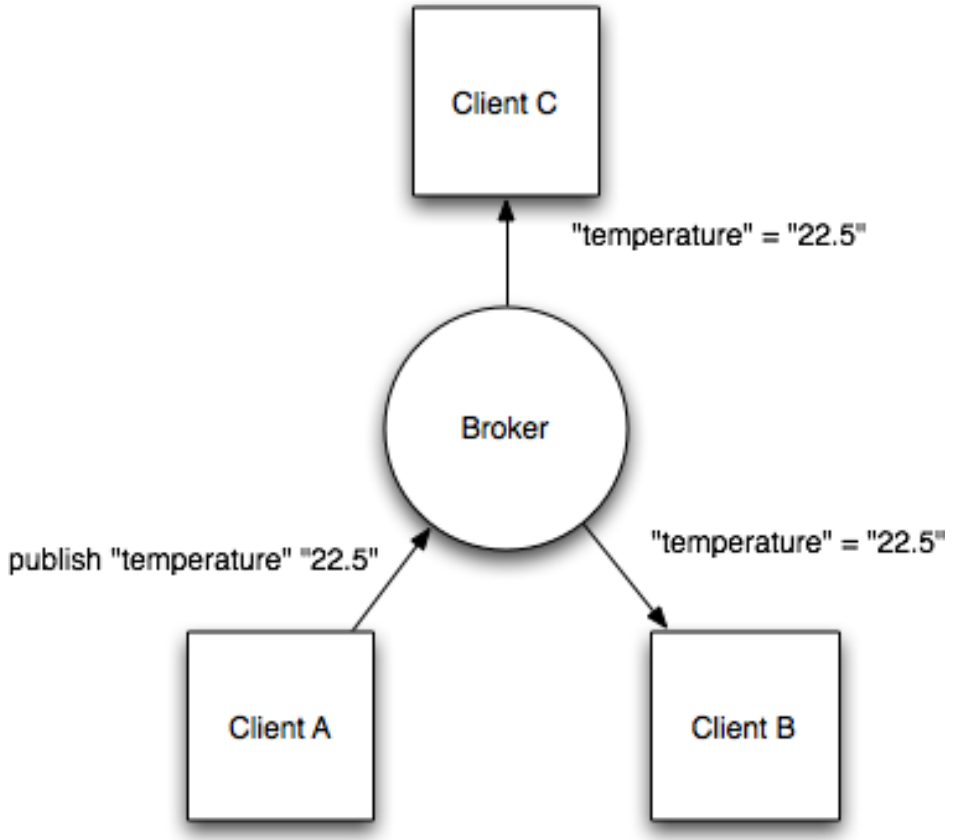
MQTT Broker

Characteristics of MQTT

- It is a machine-to-machine protocol. (M2M)
- It is a simple messaging protocol designed for constrained devices and with low bandwidth.
- It provides faster data transmission, like how WhatsApp/messenger provides a faster delivery. It's a real-time messaging protocol.
- The latest version of MQTT (Version 5.0) has the feature of enhancement in the scalability and the large-scale system to set up with thousands or millions of devices. (IoT)

MQTT Architecture

- To understand the MQTT architecture, we first look at the components of the MQTT:
- – Message
- – Client
- – Server or Broker
- – TOPIC



Contd.

- The **message** is the data carried out by the protocol across the network for the application.
- A device is a **client** if it opens the network connection to the server, publishes messages that other clients want to see, subscribes to the messages that it is interested in receiving, unsubscribes to the messages that it is not interested in receiving, and closes the network connection to the server.
- The device or a program that allows the client to publish the messages and subscribe to the messages. (**Broker**)
- The label provided to the message is checked against the subscription known by the server known as **Topic**.

Eclipse Mosquitto

- Eclipse Mosquitto is an open-source (EPL/EDL licensed) message broker that implements the MQTT protocol versions 5.0, 3.1.1, and 3.1.
- Mosquitto is lightweight and is suitable for use on all devices, from low-power single-board computers to full servers.
- To download https://mosquitto.org/download/

Basics of Mosquitto

- To start the mosquito broker, we have to type
- ***mosquitto –v***
- To see where and at what port the service is running, we type
- ***netstat –a***
- To provide a different port than the regular 1883 port
- ***mosquitto –p 4567***

Sub and PuB

- We have two roles in this system i.e. ***mosquito_sub*** and ***mosquito_pub***
- ***Sub*** is used to subscribe.
- ***Pub*** is used for publish.

To Publish

- ***mosquitto_pub –m “Hello” –t house/bulb***
- ***mosquitto_pub –m “Hello” –t house/bulb –d***

Contd.

- -h is used for help.
- -t is for the topic.
- -m is for the message.
- -d is for debugging.
- -u to provide a username.
- -P to Provide a password.
- -i to Provide client name.

Contd.

mosquitto_pub –h localhost –m “Test 1 2 3” –t house/bulb2 –u vipul –P pass –d

We have set usernames and passwords.

mosquitto_pub –h localhost –m “Test 1 2 3” –t house/bulb2 –i testid –d

We have added a test id.

To subscribe

mosquitto_sub –t iot/test

To check this open a cmd for publisher ***mosquito_pub –t iot/test –m “Hey”***

mosquitto_sub –h localhost –t house/# -C 3 -v

To see all the subscribers in the tree, we use # and -C is for no of the client. –v is for verbose it will display the topic name.

Filtering

mosquitto_sub –t iot/# -T iot/bulb2

mosquitto_pub –t iot/ -m “No”

-T is for filtering out certain groups

Assignment

- To Interface DHT11 using Thinkspeak.

Thanks